Ground Moving Target Detection With Nonuniform Subpulse Coding in SAR System

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***Abstract*— For the high-resolution and wide-swath (HRWS) synthetic aperture radar (SAR) system, the increasing imaging width results in a serious range ambiguity problem, which affects the performance of ground moving target indication (GMTI). In this article, a novel nonuniform subpulse coding (NSPC) scheme is proposed. It is characterized by resorting to range-frequency band resources and detailed coding design for each subpulse, enabling the beam auto-scanning in elevation. Also, the bandpass filtering and digital beamforming (DBF) technology with improved data reconstruction are utilized to realize the separation of subpulses and suppress range ambiguity. The NSPC technique exchanges the signal bandwidth for increasing swath without range ambiguity, and the coded subpulses can be directed to the prescribed regions, while skipping the invalid areas where the echoes are blocked. After that, through the robust principal component analysis (RPCA) method, the moving target detection is performed for each separated region without residual range-ambiguous interference. The proposed approach has been theoretically deduced in detail and the simulation experiments demonstrate its effectiveness.**

***Index Terms*— Digital beamforming (DBF), ground moving target indication (GMTI), high-resolution and wide-swath (HRWS), nonuniform subpulse coding (NSPC), range ambiguity suppression, robust principal component analysis (RPCA).**

I. Introduction

As THE theoretical researches and practical applications of synthetic aperture radar (SAR) become more and more mature [1], [2], [3], [4], the future SAR system will put forward higher requirements in terms of high-resolution and wide-swath (HRWS) imaging [5], [6], [7], ground moving target indication (GMTI) [8], [9], multimode processing [10], [11], and so on. While the SAR resolution capability is further improved, it is increasingly urgent to widen the swath. Therefore, HRWS is one of the hotspots of current research. It can provide more accurate resolution and broad field coverage ability for ground target monitoring, battlefield reconnaissance, environment detection, and terrain mapping [12], [13].

In traditional SAR systems, high pulse repetition frequency (PRF) is required for azimuth sampling unambiguously and provides fine azimuth resolution. At the same time, the radar system usually chooses a low PRF to ensure no range ambiguities in the observed swath. Owing to the limitation of minimum antenna size [14], [15], the improvement of swath width is at the expense of azimuth resolution. There has always been an irreconcilable contradiction between HRWS, and it cannot be possible to have them both. With the increase of the swath, the echoes from the previous pulse and the current pulse are received by the radar simultaneously. The echoes will overlap in the time domain, and the targets from distinct range regions will not be separated, resulting in range ambiguity [16]. In the elevation antenna pattern, strong scatters echoes entering from sidelobes can cause severe range ambiguities. Range ambiguities lead to aliasing of targets from different range regions, and will also cause clutter superimposed from multiple range ambiguities to interfere with desired signals, which are particularly detrimental to subsequent target detection and imaging. Consequently, it is crucial to tackle the range ambiguity problem in SAR-GMTI system.

In non-side-looking situations, clutter has range-dependent properties, which will bring about the broadening of the clutter Doppler spectrum. Moreover, due to the wide irradiation area, the characteristics of scatters and clutter distribution in different range regions are various. However, in traditional space-time adaptive processing (STAP), the covariance matrix is directly estimated from the training samples [17], [18], [19] of clutter after aliasing from different range ambiguities. The clutter samples do not meet the independent and identically distributed (IID) condition [20], [21], [22] under this circumstance, which will seriously deteriorate the clutter suppression performance. Clutter suppression is one of the essentials to implement effective GMTI. The direct uses of traditional STAP, displaced phase center antenna (DPCA) [23], and other methods for SAR echoes will seriously degrade the performance of moving target detection when range ambiguity and range dependence coexist. Also, since the moving targets in different range ambiguous regions may be sorely defocused, the signal-to-clutter-plus-noise ratio (SCNR) would decrease significantly. Accordingly, the slow-moving

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targets may be submerged when competing with these mixed clutters.

In order to solve the problem of range ambiguity suppression, researchers mostly focus on beampattern controlling in elevation and waveform modulation in range dimension. In [24], the range ambiguity suppression for point targets and distributed targets is achieved by combining the up and down chirp modulated waveform and the azimuth phase coding technique. In [25], an estimation and removal for the strong range ambiguities method is presented by exploiting the configuration of multistatic multiple elevation beams. In [26], [27], and [28], a nonlinear frequency modulated waveform with continuous piecewise linear instantaneous frequencies is designed in a joint range-azimuth modulation scheme for range ambiguities suppression. In [29], based on LuTan-1 satellite, a beampattern synthesis algorithm via sequential convex optimization is presented to suppress the strong co-polarized range ambiguities. In [30], beampattern synthesis by semidefinite relaxation method is proposed to shape the mainlobe and suppress the range ambiguity simultaneously. In [31], the researchers utilize prior information to calculate the power of range ambiguous regions, and propose a two-way elevation beampattern mask design approach. In [32], a superimposed stepped frequency (SSF) radar is proposed, which introduces the dimension linked to slant-range, and the clutter from each range region can be distinguished in the carrier frequency domain.

There are also many studies based on waveform diversity methods, which adjust the frequency difference, time delay information, and phase relationship of emitted signals for different transmitting channels. With the waveform diversity technique, the time-space-coupled information can be obtained, which is the key factor for distinguishing range ambiguities. Through the design of a frequency diversity array (FDA) based on multiple input multiple output (MIMO) array configuration, FDA-MIMO obtains much more available degrees of freedom (DOFs) than the actual number of antennas for HRWS and range ambiguity suppression. In [33], the secondary range dependence compensation based on FDA-MIMO is proposed by adjusting the array element spacing to discriminate range ambiguous clutter in the transmit spatial frequency domain. However, this method can only solve the number of range ambiguities is 2. In [34], a vertical FDA-MIMO architecture is described to separate range ambiguous clutter in the vertical spatial frequency domain, and by designing the nonadaptive filters to achieve unambiguous range echo extraction. In [35] and [36], aiming at side-looking geometry for FDA-MIMO, the researchers propose a joint range-angle-Doppler domain localization processing method, which realizes range ambiguous clutter suppression by utilizing the data independent pre-filter in range-angle plane. In [37], the multi-waveform covariance matrix reconstruction is adopted to suppress range ambiguous clutter by concretely designing orthogonal waveforms with frequency diversity. In [38], a slow-time coding FDA-MIMO is proposed to separate the transmitting signals by Doppler filtering. In [39] and [40], the extended azimuth phase coding is presented to suppress range ambiguities by distinguishing at the transmit spatial frequency domain. However, the assumption of MIMO system is to transmit ideal mutually orthogonal waveforms, which is difficult to realize in practical application [41].

To theoretically solve the aforementioned problem, researchers have proposed new forms of waveform diversity technology. They provide novel schemes to solve the problem of range ambiguities in medium and high repetition frequency and wide-area detection modes. In [42], the signal model and frequency scanning characteristics of the time diverse array (TDA) are preliminarily researched, and the mainlobe range ambiguity suppression for HRSW imaging is discussed. In [43], the time-of-echo technology of TDA is analyzed. In [44], the range ambiguous clutter suppression method combining TDA frequency domain filtering and azimuth phase coding technology is studied for the forward-looking array. In addition, some researchers propose the frequency scanning technique by transmitting multiple subpulses with different center frequencies to increase the imaging swath extension, where the subpulse beam will scan in elevation. By separating the subpulses occupying different frequency bands through bandpass filters, echoes pointing to different ranges can be extracted [45], [46], [47], [48]. Nevertheless, successive switching between elevation beams formed by subpulses will expend time resources.

In this article, a nonuniform subpulse coding (NSPC) technique is proposed to solve the problem of range ambiguity. By designing coded subpulses that occupy different frequency bands and do not overlap each other in transmitting time, beam scanning in elevation can be achieved. Then, the imaging areas with wide coverage can be obtained by a reasonable selection of signal operation parameters. In particular, the specific design of nonuniform modulation coefficient can make the subpulse beam point to the expected direction. Therefore, the pointing of subpulse can skip the blind range where the echoes are unable to be recorded. Next, according to the characteristic of the signal, the corresponding bandpass filters are designed to separate these subpulses. After that, the multifrequency data and digital beamforming (DBF) technology are combined to extract the expected signal and suppress the interference from sidelobes, which can further ensure that there are no range ambiguities in the current echoes. Finally, the already processed data will be imaged, and we choose robust principal component analysis (RPCA) method to detect moving targets in the different range regions.

This article is organized as follows. In Section II, the signal and structure configuration model of NSPC-SAR is constructed. In Section III, the characteristics of signal are analyzed, and the precise design criteria for NSPC signal operation parameters are described in detail. Then, the methods of range ambiguity suppression and echoes of subpulses separation are studied in Section IV, followed by the moving targets detection from the imaging results using the method of RPCA. Section V presents the simulation results and analysis. Finally, conclusions are drawn in Section VI.

II. Signal Model of NSPC-SAR With Moving Targets

In this section, we will study the NSPC-SAR framework and its signal model. The general observation geometry of

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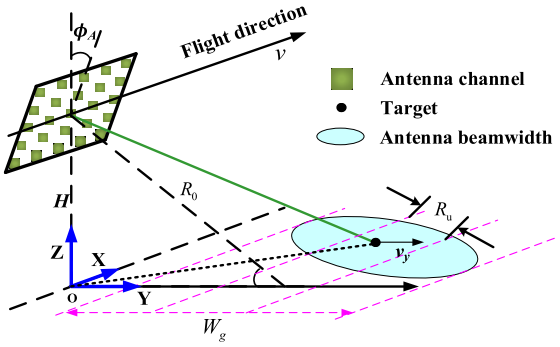

Fig. 1. Geometry of NSPC-SAR system with a moving target.

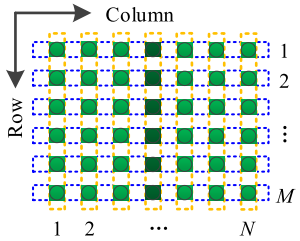

Fig. 2. Planar array configuration.

this system is shown in Fig. 1, which mainly includes a planar array and a moving target. The radar works in side-looking mode and the flight direction is along with the X-axis. The platform height and velocity are given by H and v, respectively. The total observation swath covers multiple range regions, which is represented as W_g. The planar array uniformly arranges N antenna channels at each row and M antenna channels at each column, in which the spacing between two adjacent channels is d.

In the transmitter, the waveforms in each row are duplicated and hence the entire planar array can be regarded as an M-channel linear array, where the equivalent phase centers situate at the midpoints of the rows. In the receiver, all antenna channels receive the echo signals, and the channels will be beamformed in elevation at subsequent processing. The planar array configuration has been depicted in Fig. 2.

During the coherent processing interval (CPI), the moving target keeps a constant across-track velocity on the ground v_y. Denote the slow time (azimuth time) as $t_k = k/f_{\mathrm{PRF}}$ ($k = 1, \ldots, K$), where f_{PRF} and K are PRF and the number of total pulses, respectively. At azimuth time instant $t_k = 0$, we assume that the moving target locates at $(x_0, y_0, 0)$. According to the aforementioned system configuration, the equivalent transmit array is a uniform linear array perpendicular to the horizontal plane, and the instantaneous slant-range between the mth ($m = 1, \ldots, M$) transmitting channel and the moving target can be expressed as

$$R_{t,T}^{m}(t_k) = \sqrt{\begin{array}{l}(H-(m-1)d\cos(\phi_A))^2+\left(vt_k-x_0-\frac{(N-1)d}{2}\right)^2 \\ +\left(y_0+v_y t_k+(m-1)d\sin(\phi_A)\right)^2\end{array}} \tag{1}$$

where ϕ_A is the array tilt angle in elevation. Utilizing the second-order Taylor series expansion, we obtain

$$\begin{aligned} R_{t,T}^{m}(t_k) &\approx \sqrt{R_0^2-\frac{(R_0v_r)^2}{v_e^2}+v_e^2\left(t_k+\frac{R_0v_r}{v_e^2}-\frac{x_0}{v}-\frac{(N-1)d}{2v}\right)^2} \\ &\quad +(m-1)d\sin(\phi)+v_r\frac{(N-1)d}{2v} \end{aligned} \tag{2}$$

where $R_0 = ((y_0 + v_y \cdot x_0/v) + H^2)^{1/2}$ is the slant-range at $t_k = (x_0/v)$, $v_e = ((v_y \sin(\eta_0))^2 + v^2)^{1/2}$ is the relative velocity of radar and target, $\phi = \mathrm{asin}[\sin(\phi_A) \cdot y_0/R_0 - \cos(\phi_A) \cdot H/R_0]$ is referred to as elevation angle, $\eta_0 = \mathrm{acos}((y_0 + v_y \cdot x_0/v)/R_0)$ is the grazing angle, $v_r = v_y \cos(\eta_0)$ is the velocity projected onto the slant-range plane.

The receiving array has $M \times N$ channels, and the instantaneous slant-range between the qnth ($q = 1, \ldots, M$ and $n = 1, \ldots, N$) receiving channel and the moving target can be expressed as

$$\begin{aligned} R_{t,R}^{qn}(t_k) &= \sqrt{\begin{array}{l}(H-(q-1)d\cos(\phi_A))^2+(vt_k-x_0-(n-1)d)^2 \\ +\left(y_0+v_y t_k+(q-1)d\sin(\phi_A)\right)^2\end{array}} \\ &\approx \sqrt{R_0^2-\frac{(R_0v_r)^2}{v_e^2}+v_e^2\left(t_k+\frac{R_0v_r}{v_e^2}-\frac{x_0}{v}-\frac{(n-1)d}{v}\right)^2} \\ &\quad +(q-1)d\sin(\phi)+v_r\frac{(n-1)d}{v}. \end{aligned} \tag{3}$$

At the same position, the stationary clutter point can be considered as the moving target with velocity equaling zero, i.e., $v_y = 0$, and the corresponding transmit and receive range history can be respectively represented as

$$\begin{aligned} R_{c,T}^{m}(t_k) &\approx \sqrt{R_0^2+v^2\left(t_k-\frac{x_0}{v}-\frac{(N-1)d}{2v}\right)^2} \\ &\quad +(m-1)d\sin(\phi) \end{aligned} \tag{4}$$

and

$$\begin{aligned} R_{c,R}^{qn}(t_k) &\approx \sqrt{R_0^2+v^2\left(t_k-\frac{x_0}{v}-\frac{(n-1)d}{v}\right)^2} \\ &\quad +(q-1)d\sin(\phi). \end{aligned} \tag{5}$$

The linear frequency modulated (LFM) waveform (or chirp waveform) has obvious advantages in Doppler tolerance, implementation simplicity, and high-range resolution. Therefore, the LFM waveform is employed as the basic waveform in the NSPC framework. During each pulse repetition interval (PRI), P chirp subpulses are sequentially emitted. It is worth noting that these subpulses have different centroid frequencies, and they occupy disjoint parts of the total signal bandwidth, as shown in Fig. 3. In Fig. 3, the different colors indicate different subpulses. Considering the attenuation of signal energy, in order to keep the echoes energy of different subpulses approximately the same, the subpulse duration with a farther irradiation slant-range can be designed longer. In addition, reasonable settings for the nonuniform subpulse bandwidth will be given later.

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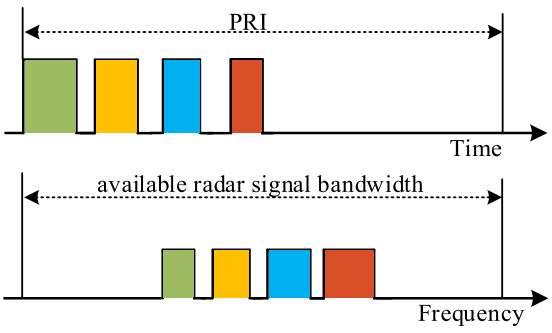

Fig. 3. Transmitting subpulses represented versus time and frequency.

Specifically, the basic waveform of the ith emitted subpulse can be formulated as

$$u_i(t) = \text{rect}\left(\frac{t-\Delta\tau_i}{T_{\text{spi}}}\right)\cdot\exp\left(j\pi\mu_i(t-\Delta\tau_i)^2\right) \tag{6}$$

where t is fast time and $\text{rect}(x)$ denotes the rectangular window function defined as

$$\text{rect}(x) = \begin{cases} 1, & |x| \le 1/2 \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

$\Delta\tau_i$ represents the time delay of the ith emitted subpulses, $i = 1, 2, \ldots, P$; μ_i is the chirp rate; T_{spi} is the subpulse duration. The design criteria of μ_i and T_{spi} for different subpulses will be given in detail in Section III-B.

In the traditional phased-array radar, these subpulses are steered in a fixed direction, which cannot achieve full-space coverage. Also, the researchers utilized the spatial weighting technique to achieve discrete elevation subpulses scanning, which requires fast phase switching. On account of giving consideration to both spatial coverage and engineering implementation efficiency, we specially design a series of coding weights, i.e., $\chi_m(i)$, for transmitting subpulses and antennas. $\chi_m(i)$ is referred to as the nonuniform modulation weight of NSPC-SAR, which is represented by

$$\begin{aligned}\chi_m(i) &= \exp(j\varphi_m(i)) \\ &= \exp\left(j\frac{2\pi\gamma_i}{M_\gamma}(m-1)(i-i_0-1)\right)\end{aligned} \tag{8}$$

where $\varphi_m(i)$ is defined as the nonuniform modulation phase, and its expression is given by

$$\varphi_m(i) = \frac{2\pi\gamma_i}{M_\gamma}(m-1)(i-i_0-1) \tag{9}$$

where M_γ is the NSPC factor which is a positive number greater than 2; γ_i is the nonuniform modulation coefficient corresponding to the ith subpulse; i_0 is a fixed number, which is designed for changing the initial beam direction.

Combining $\chi_m(i)$ and the basic waveform $u_i(t)$, one can obtain the final emitted signal for the ith subpulse and mth transmitting channel, viz.,

$$s_{m,i}(t) = u_i(t)\exp[j2\pi(f_c+\Delta f_i)(t-\Delta\tau_i)]\cdot\chi_m(i) \tag{10}$$

where f_c represents carrier frequency. The frequency difference between the ith subpulse and the first subpulse is defined as Δf_i.

With the modulation weights $\chi_m(i)$, the NSPC array radar can perform the auto-scanning of subpulses, which results in a wide-beam coverage. The detailed analysis of beampattern characteristics will be presented in Section III-A.

At the qnth receiving channel, the received target signal from the ith emitted subpulse can be described as

$$\begin{aligned}y^{\text{tar}}_{0,iqn}(t,t_k) = &\sum_{m=1}^{M}\sigma_{t,i}w_a\left(t_k+\frac{R_0v_r}{v_e^2}-\frac{x_0}{v}-\frac{(n-1)d}{2v}\right) \\ &\cdot s_{m,i}\left(t-\frac{R^m_{t,T}(t_k)+R^{qn}_{t,R}(t_k)}{c}\right).\end{aligned} \tag{11}$$

Similarly, the received clutter signal from the ith emitted subpulse can be expressed as

$$\begin{aligned}y^{\text{clu}}_{0,iqn}(t,t_k) = &\sum_{m=1}^{M}\sigma_{c,i}w_a\left(t_k-\frac{x_0}{v}-\frac{(n-1)d}{2v}\right) \\ &\cdot s_{m,i}\left(t-\frac{R^m_{c,T}(t_k)+R^{qn}_{c,R}(t_k)}{c}\right)\end{aligned} \tag{12}$$

where $\sigma_{t,i}$ and $\sigma_{c,i}$ are the scattering coefficients for target and clutter, respectively. $w_a(\cdot)$ is the azimuth window function, and c denotes the light speed.

III. Characteristics Analysis and Design Criteria for NSPC Array

A. Spatial Beampattern Characteristics Analysis

For a traditional MIMO array radar in which no coding weights exist or the weights are fixed, the mainlobes of different subpulses coincide with each other, which means that the radiation pattern does not vary with subpulses.

Conceptually, if the beampattern mainlobes of these subpulses could point to different elevation directions, one can discriminate the range of ambiguous echoes in the range frequency domain since these subpulses occupy non-overlapping parts of the total signal bandwidth. By designing the phase coding of subpulses, each subpulse can illuminate different range regions during one pulse after the synthesis of M transmitting channels. M_γ means that the elevation area covered by the entire irradiation is divided equally, but γ_i could be specifically adjusted to make the subpulses nonuniformly point to the desired directions. The specific phase coding between subpulses and transmitting channels has been shown in Fig. 4. In order to give a detailed analysis of the transmitting beampattern, we first investigate the synthetic signal for the ith subpulse at an angle ϕ, which has the form of

$$\begin{aligned}s_i(t) &= \sum_{m=1}^{M}u_i(t)\cdot\exp[j2\pi(f_c+\Delta f_i)(t-\Delta\tau_i)] \\ &\quad\cdot\exp\left(-j2\pi(m-1)\frac{d\sin(\phi)}{\lambda_i}\right)\cdot\chi_m(i) \\ &= u_i(t)P_e(\phi,i)\exp[j2\pi(f_c+\Delta f_i)(t-\Delta\tau_i)]\end{aligned} \tag{13}$$

where $\lambda_i = c/(f_c+\Delta f_i)$ is the wavelength corresponding to ith subpulse; $(\cdot)^T$ means the transpose symbol; $P_e(\phi,i)$ is the

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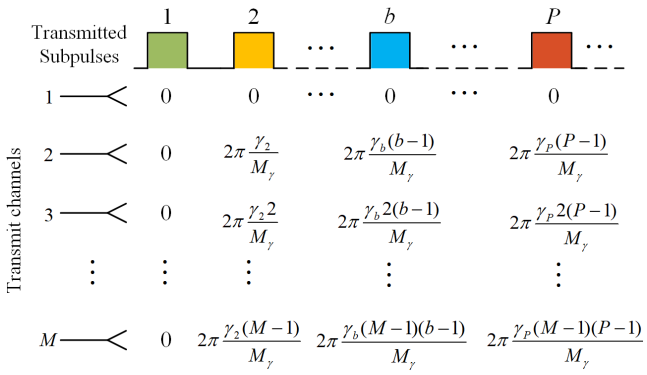

Fig. 4. Coding scheme between different subpulses and channels.

so-called transmit beampattern and the concrete expression is

$$P_e(\phi,i)=\boldsymbol{a}^T(\phi)\mathbf{d}(i)=\frac{\sin\left[M\pi\left(\frac{d\sin(\phi)}{\lambda_i}-\frac{\gamma_i(i-i_0-1)}{M_\gamma}\right)\right]}{\sin\left[\pi\left(\frac{d\sin(\phi)}{\lambda_i}-\frac{\gamma_i(i-i_0-1)}{M_\gamma}\right)\right]}\times\exp\left\{-j(M-1)\pi\left(\frac{d\sin(\phi)}{\lambda_i}-\frac{\gamma_i(i-i_0-1)}{M_\gamma}\right)\right\}. \tag{14}$$

$\boldsymbol{a}(\phi)$ and $\mathbf{d}(i)$ are the elevation spatial steering vector and coding steering vector, which can be respectively represented as

$$\boldsymbol{a}(\phi)=\left[1,\ \exp\left(j2\pi f_{a,i}(\phi)\right)\ \ldots,\exp\left(j2\pi f_{a,i}(\phi)(M-1)\right)\right]^T \tag{15}$$

$$\mathbf{d}(i)=\left[1,\ \exp\left(j2\pi f_\gamma(i)\right)\ \ldots,\exp\left(j2\pi f_\gamma(i)(M-1)\right)\right]^T \tag{16}$$

with $f_{a,i}(\phi)=d\sin(\phi)/\lambda_i$ denoting the elevation spatial frequency and $f_\gamma(i)=-\gamma_i(i-i_0-1)/M_\gamma$ denoting the coding frequency.

Obviously, with the coding steering vector, the transmitting beampointing of the array would change with subpulses. Usually, the first subpulse points to the far-end direction and the last subpulse would point to the near-end direction, which leads to a full-space beam scanning in one pulse.

Furthermore, to vividly present the propagation characteristic of the beam in space, the 2-D beampattern is depicted in Fig. 5. In Fig. 5(a), the simulation parameters are set as $M=20$, $P=100$, $M_\gamma=100$, and $\gamma_1=\gamma_2=\cdots=\gamma_P=1$, which demonstrates uniform pointing of subpulses. In addition, by adjusting the parameter P, the scope of coverage would change accordingly, as shown in Fig. 5(b), in which the relationship $P<M_\gamma$ holds. Compared with the above, the value of γ_i is no longer completely consistent, as illustrated in Fig. 5(c). Here is a sample case of nonuniform pointing, which can change the pointing direction discretely and achieve wider coverage with the same number of subpulses.

In practice, the radar beam only needs to illuminate partial space and these system parameters should be carefully designed according to the specific application requirements.

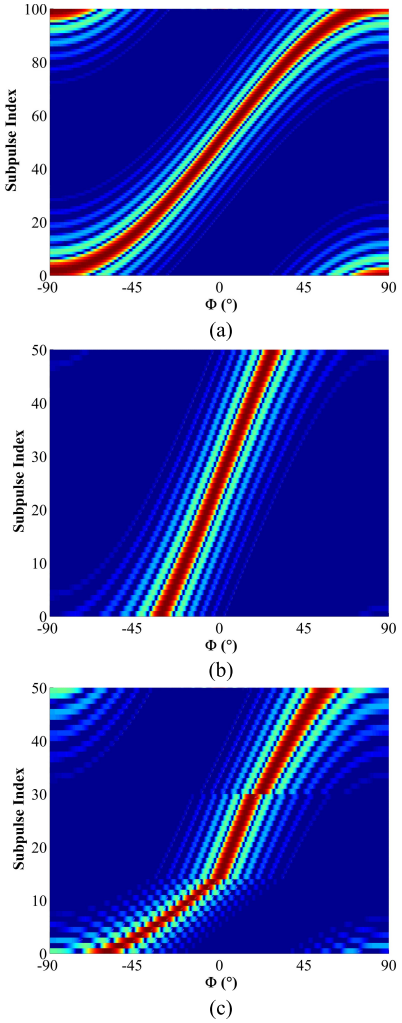

Fig. 5. Transmit beampattern comparison. (a) NSPC with $M_\gamma=P$. (b) NSPC with $M_\gamma=2P$. (c) NSPC with nonuniform pointing.

Compared with the FDA-MIMO or DBF technique, the NSPC method has more distinctive advantages. FDA-MIMO framework requires transmitting ideal orthogonal waveform. However, it is difficult to achieve in engineering. The imperfect orthogonality will lead to the cross-correlation sidelobes of waveform, which will directly affect the performance of moving target detection. In addition, the FDA application is created on the assumption that the transmit steering vector can be accurately obtained. In fact, the acquisition of the steering vector demands that the transmit waveform can be well separated, which is obviously not satisfied by the traditional MIMO waveform. By contrast, the transmit waveform of the proposed method is coherent for different channels. In other words, it can fundamentally avoid the problem of orthogonal waveform, and it is practically feasible. For receiving elevation DBF methods, they need to achieve favorable elevation angles separation ability among different range regions, so as to obtain positive performance. It is inferior to realize ambiguities separation for far-range regions by relying only on the elevation diversity. Therefore, this article resorts to the range frequency band resources to realize separation of subpulses first, and then combined with DBF technique for processing.

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B. Selection of Operation Parameters

The operation parameters of the NSPC technique would directly affect the system performance. Therefore, one should rationally select the operating parameters to ensure the full use of available resources and acquire the expected imaging and target detection performance.

Based on the basic concepts of NSPC as aforementioned, we next proceed by establishing some parameters selection criteria with concrete equations, such as the total pulse duration, subpulse number, nonuniform modulation coefficient, and so on.

First, we study the design criterion for pulse duration. In the transmitter, the entire antenna aperture is employed and the technique utilizes several subpulses to form a sequence of narrow subpulse-scanning array beams, which can illuminate the whole region of interest from far-range to near-range subswath. On the receiving end, the echoes from different subswaths are collected by all subarrays simultaneously. In this situation, these subpulse echoes would overlap in the time axis and the receive window can be greatly shortened. Therefore, the pulse echo duration can be reduced and the transmit pulse duty cycle can be improved, which can significantly decrease the radar peak power when the average power remains unchanged. At the same time, compared with the conventional SAR system, the data amount would decrease dramatically since the echo window length is cut down.

Second, the NSPC factor M_γ should be carefully designed. Ignoring the effect of scan angle, the mainlobe beamwidth of the transmitting array in elevation can be simply represented as

$$\Delta\phi_{\mathrm{BW}} = \gamma_{\mathrm{el}}\frac{\lambda_i}{M \cdot d} \tag{17}$$

where γ_{el} is a constant coefficient related to such factors as the array type, taper, and so on.

As a result, the corresponding elevation spatial frequency bandwidth is given by

$$B_{\mathrm{el}} = f_{a,i}\left(\frac{\Delta\phi_{\mathrm{BW}}}{2}\right) - f_{a,i}\left(-\frac{\Delta\phi_{\mathrm{BW}}}{2}\right) \approx \frac{1}{M}. \tag{18}$$

For the arrangement of the beam position, the elevation main beam should continuously sweep several subswaths. That is to say, the following constraint should be satisfied:

$$\frac{1}{M_\gamma} \le B_{\mathrm{el}} \Rightarrow M_\gamma \ge M. \tag{19}$$

At the same time, considering the allocation of system resources, a proper value of M_γ is M. Concretely, when the value of M_γ increases, there will be a growing overlap between two adjacent beams.

In addition, when M_γ is determined, the instantaneous direction for each subpulse can also be chosen. According to (14), the ith subpulse would point to the elevation angle ϕ_i, that is,

$$\phi_i = \arcsin\left[\frac{\lambda_i}{d}\left(z + \frac{\gamma_i(i - i_0 - 1)}{M_\gamma}\right)\right] \tag{20}$$

where z is an integer.

Also, the reflected echo signal of the pulse front edge from the far range is recorded as R_{far}, pulse back edge from the near range is R_{near}, the elevation angles corresponding to R_{far} and R_{near} are marked as ϕ_{far} and ϕ_{near}, and corresponding to *the* first subpulse and Pth subpulse are recorded as ϕ_1 and ϕ_P, respectively. To guarantee effective coverage, the observed scene should be included in the beam extent of subpulses, which can be expressed as

$$\phi_1 \le \phi_{\mathrm{far}} < \phi_{\mathrm{near}} \le \phi_P. \tag{21}$$

Substituting (20) into (21), one can obtain

$$P \ge \frac{M_\gamma d}{\lambda_P}(\sin(\phi_{\mathrm{near}}) - \sin(\phi_{\mathrm{far}})) + 1. \tag{22}$$

The relationship shown in (22) gives the selection criteria for the number of emitted subpulses.

As aforementioned, the NSPC technique can distinguish echoes from different subpulses in the range-frequency domain. The subpulses are steered to different elevation angles according to the scanning interval, which is nearly related to $1/M_\gamma$. If two range ambiguities are illuminated by the same elevation mainlobe beam of the same subpulse, the NSPC array would fail to separate them. That is to say, the smallest elevation spatial frequency difference of two range ambiguities should be larger than that of the scanning interval. Usually, the elevation spatial frequency bandwidth takes the minimum value at the far-range elevation angle. Denote $\phi_{\mathrm{far}}^{(+1)}$ as the far-end first-order ambiguity corresponding to $R_{\mathrm{far}} + R_u$, where $R_u = c/(2f_{\mathrm{PRF}})$ is the maximum unambiguous range determined by f_{PRF}. Then the following criterion for avoiding the mainlobe range ambiguities should be satisfied, viz.,

$$\frac{d}{\lambda_1}\left(\sin(\phi_{\mathrm{far}}) - \sin\left(\phi_{\mathrm{far}}^{(+1)}\right)\right) \ge \frac{1}{M_\gamma}. \tag{23}$$

Third, we will give the design criterion of the nonuniform modulation coefficient γ_i. In practical engineering implementation, the echoes will not be recorded when they are within the time of subsequent pulse transmitting, which results in the inability to receive signals in specific imaging areas, thus forming the blind ranges. Further, through the rational design of γ_i, the characteristic of nonuniform pointing can be fully utilized. This means that the beams can be pointed to skip the blind ranges in the subpulses elevation scanning, avoiding the ineffective use of time and energy resources.

Assume that the imaging area corresponds to the elevation angle $\phi \in [\phi_{\mathrm{far}}, \phi_{\mathrm{near}}]$, the corresponding slant-range is $R' = H/\sin(\phi_A + \phi)$, and it can be obtained the number of range ambiguity $N_{\mathrm{amb}} = \lfloor R'/R_u \rfloor$, where $\lfloor\cdot\rfloor$ represents the floor operation. When the slant-range satisfies the inequality $2(R' - N_{\mathrm{amb}}R_u)/c < T_p$, it is considered that the blind range occurs, where T_p is an entire pulse duration. If there exists the lth ($l = 1, \ldots, L$) certain blind range corresponding to the elevation angle $\phi_l \in [\phi_{l,\mathrm{far}}, \phi_{l,\mathrm{near}}]$, where $\phi_{l,\mathrm{far}}$ and $\phi_{l,\mathrm{near}}$ are the elevation angles corresponding to the far-end and near-end boundaries of the blind range, respectively. The following are the designs for the nonuniform coefficient γ_i that can skip this blind range and point to a specific direction.

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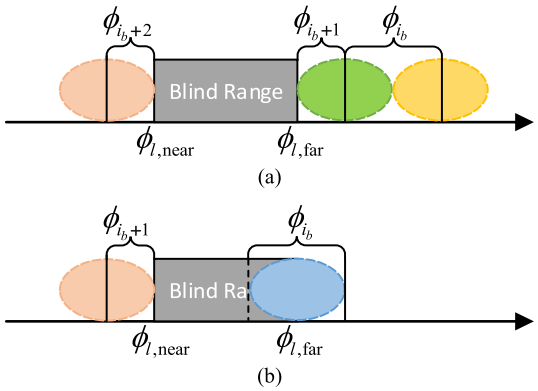

Fig. 6. Design criteria in two cases. (a) No blind range boundary included. (b) Blind range boundary included.

Due to the existence of nonuniform modulation, when γ_i of the first several subpulses have been selected, it is easy to calculate the coverage area corresponding to the mainlobe of each subpulse. Furthermore, assume that the i_bth subpulse mainlobe beam is closest to the far-end boundary of the current lth blind range. We will consider two cases according to whether the mainlobe of the i_bth subpulse contains the blind range far-end boundary. As shown in Fig. 6, the design criteria for two cases are given, where the abscissa is the elevation angle corresponding to the imaging position under the model constructed in this article.

1) No Blind Range Boundary Included: In this case, the far-end boundary of the blind range has not yet been illuminated by the mainbeam of the i_bth subpulse. Take advantage of the elevation spatial frequency to express the subpulse pointing, where $1/(2M)$ represents the spatial frequency bandwidth corresponding to the half beamwidth. When the elevation angle corresponding to the nonuniform coefficient γ_{i_b} of the i_bth subpulse satisfies

$$\frac{1}{2M} \le \frac{d\sin(\phi_{l,\text{far}})}{\lambda_{i_b}} - \frac{d\sin(\phi_{i_b})}{\lambda_{i_b}} \le \frac{3}{2M} \tag{24}$$

it can be judged as the first case, as shown in Fig. 6(a). Where ϕ_{i_b} and λ_{i_b} are the elevation angle and wavelength corresponding to the i_bth subpulse, respectively.

In order to adjust the pointing to avoid the blind range, it indicates that the $i_b + 1$th subpulse needs to be supplemented at the far-end, and the $i_b + 2$th subpulse should be guided to the near-end boundary to skip the blind range. To ensure the coverage of the $i_b + 1$th and $i_b + 2$th subpulses, the design has the following constraints:

$$0 \le \frac{d\sin(\phi_{l,\text{far}})}{\lambda_{i_b}} - \frac{d\sin(\phi_{i_b+1})}{\lambda_{i_b+1}} \le \frac{1}{2M} \tag{25}$$

and

$$0 \le \frac{d\sin(\phi_{i_b+2})}{\lambda_{i_b+2}} - \frac{d\sin(\phi_{l,\text{near}})}{\lambda_{i_b+2}} \le \frac{1}{2M} \tag{26}$$

where ϕ_{i_b+1} and λ_{i_b+1} are the elevation angle and wavelength corresponding to the $i_b + 1$th subpulse, and ϕ_{i_b+2} and λ_{i_b+2} are corresponding to the $i_b + 2$th subpulse.

Taking (20) into different elevation angles, according to the inequality conditions of (25) and (26), the design criteria can be obtained as

$$\gamma_{i_b+1} \in \left[\left(\frac{d\sin(\phi_{l,\text{far}})}{\lambda_{i_b}} - \frac{1}{2M}\right)\frac{M_\gamma}{(i_b - i_0)}, \frac{M_\gamma d\sin(\phi_{l,\text{far}})}{\lambda_{i_b}(i_b - i_0)}\right] \tag{27}$$

and

$$\gamma_{i_b+2} \in \left[\frac{M_\gamma d\sin(\phi_{l,\text{near}})}{\lambda_{i_b+2}(i_b - i_0 + 1)}, \frac{\left(2Md\sin(\phi_{l,\text{near}}) + \lambda_{i_b+2}\right)M_\gamma}{2M\lambda_{i_b+2}(i_b - i_0 + 1)}\right] \tag{28}$$

where γ_{i_b+1} and γ_{i_b+2} are the nonuniform coefficients corresponding to the $i_b + 1$th and $i_b + 2$th subpulses, respectively.

2) Blind Range Boundary Included: When the coverage of the i_bth subpulse has already contained the far-end boundary, that is,

$$\left|\frac{d\sin(\phi_{l,\text{far}})}{\lambda_{i_b}} - \frac{d\sin(\phi_{i_b})}{\lambda_{i_b}}\right| \le \frac{1}{2M} \tag{29}$$

it can be judged as the second case, as shown in Fig. 6(b).

At this time, it is only necessary to directly ensure that the $i_b + 1$th subpulse skips the blind range, and the constraint is expressed as

$$0 \le \frac{d\sin(\phi_{i_b+1})}{\lambda_{i_b+1}} - \frac{d\sin(\phi_{l,\text{near}})}{\lambda_{i_b+2}} \le \frac{1}{2M}. \tag{30}$$

By solving (30), we can get the design criterion is

$$\gamma_{i_b+1} \in \left[\frac{M_\gamma d\sin(\phi_{l,\text{near}})}{\lambda_{i_b+1}(i_b - i_0)}, \frac{M_\gamma\left(\lambda_{i_b+1} + 2Md\sin(\phi_{l,\text{near}})\right)}{2M\lambda_{i_b+1}\lambda_{i_b}(i_b - i_0)}\right]. \tag{31}$$

Finally, we will give the criteria for subpulse frequency increment Δf_i in terms of consistent ground resolution. We know that the range resolution is determined by the transmitting signal bandwidth, which can be expressed as $\rho_{ri} = c/(2B_i)$, where ρ_{ri} is the range resolution, and $B_i = \mu_i T_{\text{spi}}$ is signal bandwidth corresponding to the ith subpulse. According to the geometric configuration of the scene, the relationship between ground resolution and range resolution can be written as

$$\rho_{wi} = \frac{\rho_{ri}}{\cos(\phi_i + \phi_A)} = \frac{c}{2B_i\cos(\phi_i + \phi_A)} \tag{32}$$

where ρ_{wi} is the ground resolution of the ith subpulse. For the sake of composing the final imaging result has the same ground resolution in each subpulse imaging area, that is, to satisfy $\rho_{w1} = \rho_{w2} = \cdots = \rho_w$, where ρ_w is the desired same ground resolution. Therefore, the bandwidth of the subpulse should be changed accordingly, and the bandwidth can be produced as

$$B_i = \frac{c}{2\rho_w\cos(\phi_i + \phi_A)}. \tag{33}$$

This means that only a relatively small bandwidth is required for the far-range scene, while a slightly larger bandwidth is chosen for the near-range circumstance. If the attenuation of

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signal energy with distance is considered, it can be further designed to increase the subpulse duration T_{spi} corresponding to the far-range condition and reduce the chirp rate μ_i at the same time.

Regarding the selection of subpulse frequency increment, under simplified processing, the widest subpulse bandwidth is selected as the frequency interval of all subpulses. In this manner, the bandwidth resources would be wasted. The corresponding frequency increment is selected as

$$\Delta f_i \geq B_{\max} = \frac{c}{2\rho_w \cos(\phi_{\text{near}} + \phi_A)} \tag{34}$$

where $B_{\max}$ is the widest subpulse bandwidth. To achieve the maximum utilization of the bandwidth resources, the frequency increment for these subpulses can be designed as

$$\begin{aligned} &\Delta f_i - \Delta f_{i-1} \\ &\geq \frac{1}{2}\left(\frac{c}{2\rho_w \cos(\phi_i + \phi_A)} + \frac{c}{2\rho_w \cos(\phi_{i-1} + \phi_A)}\right) + \Delta f_e \\ &\qquad i = 2, \ldots, P \end{aligned} \tag{35}$$

where Δf_e is an additional frequency offset to ensure the frequency band discrimination.

IV. Ambiguity Suppression of NSPC-SAR Echoes

A. Range Ambiguity Suppression and Echo Separation in Frequency Domain

After collecting the echoes of these subpulses, one should accomplish the waveform separation and range ambiguity suppression before subsequent moving target detection.

In the following, we take the target signal as an example of formula derivation. First, in order to obtain the ith subpulse signal, one should down-convert the measured echoes with the corresponding function, which can be given by

$$H_d^{(i)}(t) = \exp[j2\pi (f_c + \Delta f_i)t]. \tag{36}$$

Multiplying this down-conversion function with the mixed echoes, the obtained results of ith subpulse and other subpulses can be respectively expressed as

$$\begin{aligned} &y_{iqn}^{\text{tar}}(t, t_k) \\ &= \sum_{m=1}^{M} \sigma_{t,i} w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right) \\ &\quad \cdot u_i\left(t - \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right) \cdot \chi_m(i) \\ &\quad \cdot \exp\left[-j2\pi (f_c + \Delta f_i)\left(\Delta\tau_i + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \end{aligned} \tag{37}$$

and

$$\begin{aligned} &y_{\text{other},qn}^{\text{tar}}(t, t_k) \\ &= \sum_{h=1, h\neq i}^{P} \sum_{m=1}^{M} \sigma_{t,h} w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right) \cdot \chi_m(h) \\ &\quad \cdot u_h\left(t - \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right) \cdot \exp(j2\pi (\Delta f_h - \Delta f_i)t) \\ &\quad \cdot \exp\left[-j2\pi (f_c + \Delta f_h)\left(\Delta\tau_h + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \end{aligned} \tag{38}$$

where h ($h = 1, \ldots, P$ and $h \neq i$) is the index of other subpulses.

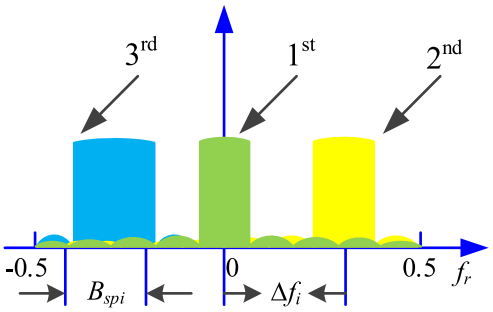

Fig. 7. Range frequency spectra distribution of the subpulses.

Performing the range Fourier transform to $y_{iqn}^{\text{tar}}(t, t_k)$ and $y_{\text{other},qn}^{\text{tar}}(t, t_k)$ yields

$$\begin{aligned} &Y_{iqn}^{\text{tar}}(f_r, t_k) \\ &= \int y_{iqn}^{\text{tar}}(t, t_k) \cdot \exp(-j2\pi f_r t) dt \\ &= \sum_{m=1}^{M} \sigma_{t,i} w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right) \\ &\quad \cdot \text{rect}\left(\frac{f_r}{B_{\text{spi}}}\right) \cdot \exp\left(-j\pi \frac{f_r^2}{\mu_i}\right) \cdot \chi_m(i) \\ &\quad \cdot \exp\left[-j2\pi (f_r + f_c + \Delta f_i)\left(\Delta\tau_i + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \end{aligned} \tag{39}$$

and

$$\begin{aligned} &Y_{\text{other},qn}^{\text{tar}}(f_r, t_k) \\ &= \sum_{h=1, h\neq i}^{P} \sum_{m=1}^{M} \sigma_{t,h} w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right) \cdot \chi_m(h) \\ &\quad \cdot \text{rect}\left(\frac{f_r - \Delta f_h + \Delta f_i}{B_{\text{sph}}}\right) \cdot \exp\left(-j\pi \frac{(f_r - \Delta f_h + \Delta f_i)^2}{\mu_h}\right) \\ &\quad \cdot \exp\Bigg(-j2\pi (f_r - \Delta f_h + \Delta f_i) \\ &\qquad \times \left(\Delta\tau_h + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\Bigg) \\ &\quad \cdot \exp\left[-j2\pi (f_c + \Delta f_h)\left(\Delta\tau_h + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \end{aligned} \tag{40}$$

where $f_r \in [-B_{\text{spi}}/2, B_{\text{spi}}/2]$ is the range frequency variable.

Concretely, we take three subpulses as an example and the frequency spectra are sketched in Fig. 7. Obviously, the main part of an arbitrary subpulse is isolated from that of adjacent subpulses. However, some residual out-of-band interference energies are still mixed with each other.

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What follows is the separation of subpulses. One can accomplish the subpulse extraction by the bandpass filtering operation. After the down-conversion procedure with $H_d^{(i)}(t)$, the main body of the frequency variety extent of the ith subpulse is limited to $[-B_{\text{spi}}/2, B_{\text{spi}}/2]$.

In this case, we can construct the bandpass filter as

$$H_{bp}^{(i)}(f_r) = \begin{cases} 1, & f_r \in \left[-B_{\text{spi}}/2, B_{\text{spi}}/2\right] \\ 0, & \text{otherwise.} \end{cases} \tag{41}$$

Utilizing the filter $H_{bp}^{(i)}(f_r)$ to segment the expected parts from the total frequency spectra yields the frequency component of the ith subpulse echo. The next step is to accomplish the range compression with the corresponding matched filter, which has the form of

$$H_{\text{match}}^{(i)}(f_r) = \text{rect}\left(\frac{f_r}{B_{\text{spi}}}\right)\exp\left(j\pi\frac{f_r^2}{\mu_i}\right). \tag{42}$$

Multiplying the two filters $H_{bp}^{(i)}(f_r)$ and $H_{\text{match}}^{(i)}(f_r)$ with the mixed echoes in the frequency domain, i.e., $Y_{iqn}^t + Y_{\text{other},qn}^t$, and then transforming the obtained results into the time domain, the extracted echo signal of ith subpulse can be acquired, that is,

$$\begin{aligned} &y_{iqn}^{\text{tar}}(t, t_k) \\ &= \sum_{m=1}^{M} \bar{\sigma}_{t,i}\, w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right)\cdot \chi_m(i) \\ &\quad \cdot\text{sinc}\left[B_{\text{spi}}\left(t - \Delta\tau_i - \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \\ &\quad \cdot\exp\left[-j2\pi\left(f_c + \Delta f_i\right)\left(\Delta\tau_i + \frac{R_{t,T}^m(t_k) + R_{t,R}^{qn}(t_k)}{c}\right)\right] \end{aligned} \tag{43}$$

where $\bar{\sigma}_{t,i}$ is the amplitude after range compression.

B. Residual Range Ambiguity Suppression in Spatial Domain

Through the frequency domain separation, the majority of the ambiguous energies and interferences from adjacent subpulses have been eliminated. However, the sidelobe ambiguous energies of the current subpulse will also interfere with the expected signal of the mainlobe. In addition, some residual interferences from other subpulses after bandpass filtering still remain in the currently extracted subpulse echoes. To deal with these issues, we will implement the secondary range ambiguity suppression in the spatial domain.

Ignoring the channel envelope changes in the planar array, the equation in (43) can be simplified as

$$y_{iqn}^{\text{tar}}(t, t_k) = y_{\text{in,ref}}^{\text{tar}}(t, t_k)\cdot\exp\left(-j2\pi(q-1)\frac{d}{\lambda_i}\sin(\phi)\right) \tag{44}$$

and

$$\begin{aligned} y_{\text{in,ref}}^{\text{tar}}(t, t_k) &= \bar{\sigma}_{t,i}\, w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right) \\ &\quad \cdot\text{sinc}\left[B_{\text{spi}}\left(t - \Delta\tau_i - \frac{2R_t^n(t_k)}{c}\right)\right]\cdot P_e(\phi, i) \\ &\quad \cdot\exp\left[-j2\pi\left(f_c + \Delta f_i\right)\left(\Delta\tau_i + \frac{2R_t^n(t_k)}{c}\right)\right] \end{aligned} \tag{45}$$

where

$$\begin{aligned} &R_t^n(t_k) \\ &\approx \sqrt{R_0^2 - \frac{(R_0 v_r)^2}{v_e^2} + v_e^2\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right)^2} \\ &\quad + v_r\frac{(n-1)d}{2v} \end{aligned} \tag{46}$$

represents the instantaneous slant-range between the reference point in the nth column and the moving target. Considering the nth column of the receive array, which has M channels in elevation, the steering vectors for the ith subpulse and other subpulses can be, respectively, formulated as

$$\begin{aligned} \boldsymbol{a}_i(\phi_i) = \big[&1,\ \exp\left(j2\pi f_{a,i}(\phi_i)\right) \\ &\ldots, \exp\left(j2\pi f_{a,i}(\phi_i)(M-1)\right)\big]^T \end{aligned} \tag{47}$$

and

$$\begin{aligned} \boldsymbol{a}_h(\phi_h) = \big[&1,\ \exp\left(j2\pi f_{a,h}(\phi_h)\right) \\ &\ldots, \exp\left(j2\pi f_{a,h}(\phi_h)(M-1)\right)\big]^T \end{aligned} \tag{48}$$

where $f_{a,h}(\phi_h) = d\sin(\phi_h)/\lambda_h$ and $\lambda_h = c/(f_c + \Delta f_h)$, $\phi_h = \arcsin[(\lambda_h/d)(z + (\gamma_h(h - i_0 - 1)/M_\gamma))]$ is the beampointing of the hth subpulse.

Stacking the echoes of the M channels in the nth column, the corresponding data vectors of the ith subpulse and hth subpulse in the residual echoes can be, respectively, expressed as

$$\mathbf{y}_{\text{in}}^{\text{tar}}(t, t_k) = \left[y_{i1n}^{\text{tar}}(t, t_k), y_{i2n}^{\text{tar}}(t, t_k), \cdots\cdots, y_{iMn}^{\text{tar}}(t, t_k)\right] \tag{49}$$

$$\mathbf{y}_{\text{hn}}^{\text{tar}}(t, t_k) = \left[y_{h1n}^{\text{tar}}(t, t_k), y_{h2n}^{\text{tar}}(t, t_k), \cdots\cdots, y_{hMn}^{\text{tar}}(t, t_k)\right] \tag{50}$$

where $y_{hqn}^{\text{tar}}(t, t_k)$ is the residual signal of the hth subpulse after frequency domain separation. The derivation is similar to (45), and we would not explore it here. Likewise, the clutter echoes of the corresponding subpulses $\mathbf{y}_{\text{in}}^{\text{clu}}(t, t_k)$ and $\mathbf{y}_{\text{hn}}^{\text{clu}}(t, t_k)$ can be obtained when $v_y = 0$ in the expression. $\mathbf{y}_{inr}^{\text{tar}}$ and $\mathbf{y}_{hnr}^{\text{tar}}$ used in subsequent calculation are formed by stacking the data of the rth range bin from M channels, which is consistent with (49) and (50).

In the following, the DBF technique is utilized to further suppress the residual interferences and ambiguities. We process the data of range bins in turn. The optimal weighting vector can be acquired by constraining the interferences power while improving the wanted echo power, that is,

$$\begin{cases} \min & \mathbf{w}_{ir}^H\mathbf{R}_{-ir}\mathbf{w}_{ir} \\ \text{s.t.} & \mathbf{w}_{ir}^H\boldsymbol{a}_i(\phi_i) = 1 \\ & \mathbf{w}_{ir}^H\boldsymbol{a}_h(\phi_h) = 0 \end{cases} \tag{51}$$

where $\mathbf{w}_{ir}$ is the weighting vector for extracting the echo of the ith subpulse for the rth range bin, $(\cdot)^H$ represents the conjugate transpose operation, and $\mathbf{R}_{-ir}$ can be formulated as

$$\mathbf{R}_{-ir} = \sum_{h=1, h\neq i}^{P} A_{hr}^2\boldsymbol{a}_h(\phi_h)\boldsymbol{a}_h^H(\phi_h) \tag{52}$$

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where A_{hr}^2 is the power of the rth range bin in the given swath, and it can be obtained by the subpulse which is exactly pointing to the elevation angle ϕ_h. A_{hr}^2 is deduced according to the Capon spatial spectrum estimator [49], [50] and can be constructed as

$$A_{hr}^2 = \frac{1}{\boldsymbol{a}_h^H(\phi_h)\hat{\mathbf{R}}_{hr}^{-1}\boldsymbol{a}_h(\phi_h)} \tag{53}$$

where $\hat{\mathbf{R}}_{hr} = \mathbf{y}_{hnr}(\mathbf{y}_{hnr})^H$ is the rth range bin of the hth subpulse sample covariance matrix, and $\mathbf{y}_{hnr} = \mathbf{y}_{hnr}^{\text{tar}} + \mathbf{y}_{hnr}^{\text{clu}}$ is the echo data of target plus clutter.

Notably, the reconstruction of $\mathbf{R}_{-ir}$ would directly affect the subsequent clutter suppression and target detection performance. Usually, A_{hr}^2 can be calculated using the geometry relationship between the radar and observed point and the radar equation. However, this way could not represent the backscattering coefficients of the observed scene, which only contains the power information varying with slant-range. Wen et al. [37] point out that the signal powers for multifrequency data can be considered approximately equal to each other when the frequency spacing is small (compared with carrier frequency). In our NSPC method, the amplitude information of ith subpulse echo after bandpass filtering could basically reflect the backscattering coefficients for the corresponding range region. Therefore, we use this property to reconstruct the data covariance matrix $\mathbf{R}_{-ir}$. Compared with those methods only depending on the geometry relationship, the proposed scheme has advantages in accurate power information reconstruction.

Based on the minimum variance distortionless response (MVDR) criterion, the weighting coefficients can be calculated as

$$\mathbf{w}_{ir} = \mathbf{R}_{-ir}^{-1}\mathbf{C}_1\left(\mathbf{C}_1^H\mathbf{R}_{-ir}^{-1}\mathbf{C}_1\right)^{-1}\mathbf{F}_1 \tag{54}$$

where $\mathbf{C}_1 = [\boldsymbol{a}_i(\phi_i), \boldsymbol{a}_1(\phi_1), \ldots, \boldsymbol{a}_{P-1}(\phi_{P-1})] \in \mathbb{C}^{M\times P}$, and the constraint vector $\mathbf{F}_1$ is $\mathbf{F}_1 = [1, 0, 0, \ldots, 0]^T$, $\mathbf{F}_1 \in \mathbb{R}^{P\times 1}$.

With these weighting coefficients, the final extracted signal for the rth range bin of the ith subpulse at the nth receive array can be acquired, that is,

$$\hat{y}_{inr}(t, t_k) = \mathbf{w}_{ir}^H\left(\mathbf{y}_{inr}(t, t_k) + \sum_{h=1, h\neq i}^{P}\mathbf{y}_{hnr}(t, t_k)\right) \tag{55}$$

where $\mathbf{y}_{inr}(t, t_k) = \mathbf{y}_{inr}^{\text{tar}}(t, t_k) + \mathbf{y}_{inr}^{\text{clu}}(t, t_k)$. All range bins are processed circularly, and $\hat{y}_{inr}(t, t_k)$ is arranged according to the range direction to obtain the final extraction result $\hat{y}_{\text{in}}(t, t_k)$.

At this point, the M channels in elevation have been merged into one channel. Besides, since the N channels are uniformly distributed in azimuth, the clutter suppression and moving target detection can be completed with the azimuth DOFs.

C. Moving Target Detection

In this section, we first study the azimuth focusing of the echo data. With the azimuth compression operation, the signal power could be improved by obtaining the pulse accumulation gain, which is closely related to the target detection performance.

After the above subpulses extraction and range ambiguities suppression, the target signal of the nth receiving channel has the form of

$$\begin{aligned} &y_{\text{in}}^{\text{tar}}(t, t_k) \\ &= w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right)p_{ri}\left(t - \Delta\tau_i - \frac{2R_t^n(t_k)}{c}\right) \\ &\quad\cdot\exp\left[-j2\pi\,(f_c + \Delta f_i)\left(\Delta\tau_i + \frac{2R_t^n(t_k)}{c}\right)\right] \end{aligned} \tag{56}$$

where $p_{ri}(\cdot) = \text{sinc}[B_{\text{spi}}(\cdot)]$.

Transforming $y_{\text{in}}^{\text{tar}}(t, t_k)$ into the 2-D frequency domain, the expression can be written as

$$\begin{aligned} &Y_{\text{in}}^{\text{tar}}(f_r, f_a) \\ &= W_a(f_a)\cdot\text{rect}\left(\frac{f_r}{B_{\text{spi}}}\right)\cdot\exp[-j2\pi\,(f_c + f_r + \Delta f_i)\Delta\tau_i] \\ &\quad\cdot\exp\left[-j\frac{4\pi\,(f_c + f_r + \Delta f_i)}{c}\left(v_r\frac{(n-1)d}{2v}\right)\right] \\ &\quad\cdot\exp\left[j2\pi f_a\left(\frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right)\right] \\ &\quad\cdot\exp\left[-j\frac{2\pi R_0}{v_e}\sqrt{1 - \frac{v_r^2}{v_e^2}}\sqrt{f_{aM}^2 - f_a^2}\right] \\ &\quad\cdot\exp\left[-j\frac{4\pi R_0 v}{c v_e}\sqrt{1 - \frac{v_r^2}{v_e^2}}\left(\frac{f_{aM}}{\sqrt{f_{aM}^2 - f_a^2}}\right.\right. \\ &\qquad\left.\left. - \frac{v f_a^2 f_r}{c\left(f_{aM}^2 - f_a^2\right)^{3/2}}\right)f_r\right] \end{aligned} \tag{57}$$

where $W_a(\cdot)$ is the frequency domain form of the azimuth window function and $f_{aM} = (2v_e/c)(f_c + \Delta f_i)$.

According to (57), the range migration correction function can be established as

$$H_{\text{RCM}} = \exp\left[j\frac{4\pi R_0}{c}\left(\frac{1}{2}\left(\frac{f_a}{f_{aM}}\right)^2 - \frac{v f_a^2 f_r}{c\left(f_{aM}^2 - f_a^2\right)^{3/2}}\right)f_r\right]. \tag{58}$$

The azimuth compression filter function is constructed as

$$H_{\text{azi}} = \exp\left[j\frac{2\pi R_0}{v}\sqrt{f_{aM}^2 - f_a^2}\right]. \tag{59}$$

It should be noted that the unknown information of target velocity does not appear in the filter function, that is, it does not affect the construction of the filter. In this case, the inaccuracy of the velocity will bring about the mismatch and defocusing of the azimuth dimension. However, the platform velocity v is much greater than the velocity v_y of the moving target generally, so the unknown target velocity will not affect azimuth spectrum compression [51].

Perform range migration correction and azimuth compression on (57), and then convert it to the time domain, we can obtain

$$\begin{aligned} &y_{\text{in}}^{\text{tar}}(t, t_k) \\ &= w_a\left(t_k + \frac{R_0 v_r}{v_e^2} - \frac{x_0}{v} - \frac{(n-1)d}{2v}\right)p_{ri}\left(t - \Delta\tau_i - \frac{2R_0}{c}\right) \end{aligned}$$

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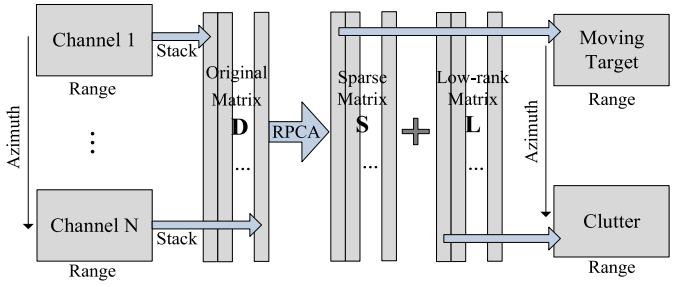

Fig. 8. Process of RPCA.

$$\cdot\exp\left[-j2\pi\,(f_c+\Delta f_i)\left(\Delta\tau_i+v_r\frac{(n-1)d}{cv}\right)\right]. \tag{60}$$

After the above azimuth compression, the focusing imaging of each subpulse echo is completed. As can be seen from (60), the moving target information is still retained in the difference of phase term related to the target velocity and the channel spacing. Consequently, this information can be analyzed to achieve moving target detection. We utilize the RPCA method, which exploits the property of the matrix rank to separate out matrices with different components. Compared with the entire observation scene, moving targets generally satisfy the sparsity, while the background clutter between different channels is similar, which makes it meet the low-rank characteristic. Therefore, the sparse matrix containing moving targets and the low-rank matrix containing clutter can be decomposed by the RPCA method. Moreover, RPCA has good moving target detection performance in strong clutter background [52], and it can be performed in the range-Doppler domain or the image domain [53]. Researchers have proposed many methods to solve the RPCA problem, and we employ the go decomposition (GoDec) method in this article. The GoDec model [54] can convert the parameters that are hard to determine in the original RPCA optimization problem into the more easily representable rank and cardinality, and the model is expressed as

$$\begin{aligned}&\min_{\mathbf{L},\mathbf{S}}\ \|\mathbf{D}-\mathbf{L}-\mathbf{S}\|_F^2\\&\text{s.t. rank}\quad(\mathbf{L})\le r,\ \text{card}(\mathbf{S})\le k\end{aligned} \tag{61}$$

where the original data matrix $\mathbf{D}$ is to vectorize the received echoes and stack the data of all channels into a matrix, as shown in Fig. 8, and the expression is

$$\mathbf{D}=[\text{vec}(y_{i1})\ \text{vec}(y_{i2})\cdots\text{vec}(y_{iN})]^T \tag{62}$$

where vec$(\cdot)$ indicates the vectorization operator, $y_{\text{in}}=y_{\text{in}}^{\text{tar}}+y_{\text{in}}^{\text{clu}}$ $(n=1,\ldots,N)$ is the echo data, and $y_{\text{in}}^{\text{clu}}$ can be acquired when $v_y=0$. $\mathbf{L}$ and $\mathbf{S}$ are the low-rank matrix and sparse matrix, respectively; rank$(\mathbf{L})$ is the rank of the low-rank matrix; card$(\mathbf{S})$ is the cardinality of the sparse matrix, and when the value of k is set to be smaller stands for the matrix $\mathbf{S}$ is more sparse; $\|\cdot\|_F$ denotes the Frobenius norm. The clutter between channels is consistent ideally, so r can be set to 1. While the value of k is actually unknown, it can be selected as slightly larger than the resolution units occupied by moving targets generally [53].

TABLE I
GODEC ALGORITHM

Algorithm GoDec
Input: $\mathbf{D}$, r, k, ε, q
Output: $\mathbf{L}$, $\mathbf{S}$
Initialize: $\mathbf{L}_0:=\mathbf{D}$, $\mathbf{S}_0:=\mathbf{0}$, $t:=0$
while $\|\mathbf{D}-\mathbf{L}_t-\mathbf{S}_t\|_F^2/\|\mathbf{D}\|_F^2>\varepsilon$ **do**
$t:=t+1$;
$\hat{\mathbf{L}}=[(\mathbf{D}-\mathbf{S}_{t-1})(\mathbf{D}-\mathbf{S}_{t-1})^T]^q(\mathbf{D}-\mathbf{S}_{t-1})$;
$\mathbf{Y}_1=\hat{\mathbf{L}}\mathbf{A}_1$, $\mathbf{A}_2=\mathbf{Y}_1$;
$\mathbf{Y}_2=\hat{\mathbf{L}}^T\mathbf{Y}_1=\mathbf{Q}_2\mathbf{R}_2$, $\mathbf{Y}_1=\hat{\mathbf{L}}\mathbf{Y}_2=\mathbf{Q}_1\mathbf{R}_1$;
If rank$(\mathbf{A}_2^T\mathbf{Y}_1)<r$ **then** $r:=$ rank$(\mathbf{A}_2^T\mathbf{Y}_1)$, go to the first step; **end**;
$\mathbf{L}_t=\mathbf{Q}_1[\mathbf{R}_1(\mathbf{A}_2^T\mathbf{Y}_1)^{-1}\mathbf{R}_2^T]^{1/(2q+1)}\mathbf{Q}_2^T$;
$\mathbf{S}_t=\mathbf{P}_\Omega(\mathbf{D}-\mathbf{L}_t)$, Ω is the nonzero subset of the
first k largest entries of $\|\mathbf{D}-\mathbf{L}_t\|$;
end while

GoDec algorithm is solved by bilateral random projection (BRP) to alternately optimize $\mathbf{L}$ and $\mathbf{S}$ [55]. Compared with the traditional singular value decomposition (SVD), this algorithm has significantly improved operation speed and can effectively decrease the computational complexity. The steps of the GoDec algorithm are shown in Table I. The data from the output $\mathbf{L}$ and $\mathbf{S}$ are taken out in columns and rearranged into the matrix. In other words, the sparse matrix representing the moving target is extracted from the clutter background, and the corresponding moving target detection result is obtained.

The main steps of range ambiguity suppression for the proposed NSPC method are summarized in Table II.

V. SIMULATION AND NUMERICAL RESULTS

In this section, the NSPC strategy is applied to point targets and distributed targets respectively to verify the performance of range ambiguity suppression. The system design parameters are presented in Table III. In the simulation, the number of subpulses is set to 3, occupying different frequency bands, and precisely designed to point to three specific range ambiguous regions in turn. The grazing angles of these three regions are concretely 5.9°, 10.4°, and 47°, and the corresponding nonuniform modulation coefficients γ_i are calculated according to (20) as 3.71, 5.11, and 0.84, respectively. In the transmitter, we use the entire planar array, namely, 32 rows and 6 columns of antennas, to transmit signals. In the receiver, the method of subarray receiving is adopted. In elevation, four receiving channels are utilized for residual range ambiguity suppression by DBF, and in azimuth, three receiving channels are set for moving target detection. In addition, the azimuth antenna aperture is set to 2 m.

A. Simulation of Point Targets for NSPC-SAR

In this section, we will demonstrate the range ambiguity suppression performance of the proposed scheme using the

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TABLE II
MAIN STEPS OF THE PROPOSED NSPC FRAMEWORK

Design the NSPC signal according to Section III-B and transmit the waveform in (10).
Receive radar echo data and **Repeat**

Step 1) Select to extract the *i*th subpulse, perform the down-conversion process with (36), and transform to range-frequency domain to get (39)-(40).

Step 2) Adopt the bandpass filter with (41), complete the range compression by (42), and convert to 2-D time domain as shown in (43).

Repeat

Step 3) Take the *r*th range bin data of *M* row channels for stacking, as shown in (49)-(50), and use the stacked data to obtain the covariance matrix.

Step 4) Combine the steering vectors in (47)-(48) with the covariance matrix, and obtain the weights for the *r*th range bin by (52)-(54).

Step 5) Multiply the weights with the *r*th range bin data to get the unambiguous results, as shown in (55).

Until the data of all range bins are processed.

Step 6) Transform to 2-D frequency domain in (57), perform migration correction and azimuth compression according to (58)-(59), and obtain imaging result in (60).

Step 7) Implement vectorization of *N* column channels with (62), and realize detection by Table I algorithm.

Until all subpulses are processed.

TABLE III
RADAR SYSTEM PARAMETERS

Parameters	Values	Parameters	Values
Carrier Frequency	18GHz	***PRF***	1200Hz
Platform Velocity	500m/s	***Platform Height***	30km
Array Tilt Angle	50°	***Transmit Channel Spacing***	0.0083m
Subpulse Number	3	***Subpulse Width***	3μs
Subpulse Bandwidth	37.5MHz	***NSPC factor***	32

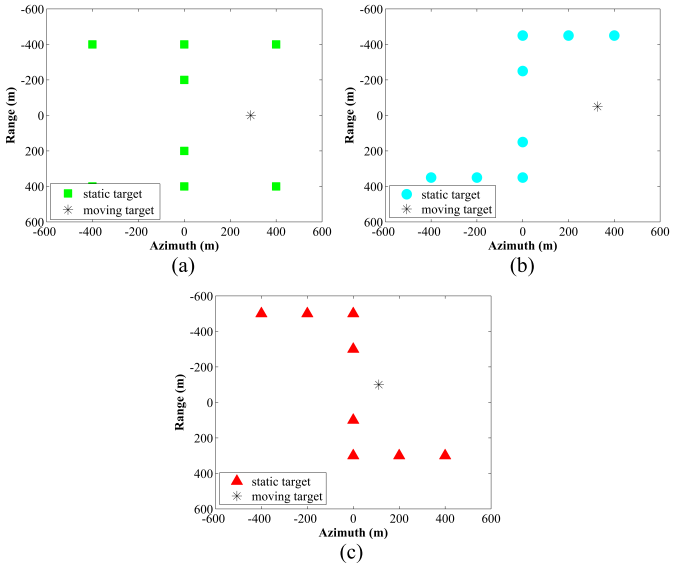

Fig. 9. Targets positions of three regions. (a) First region. (b) Second region. (c) Third region.

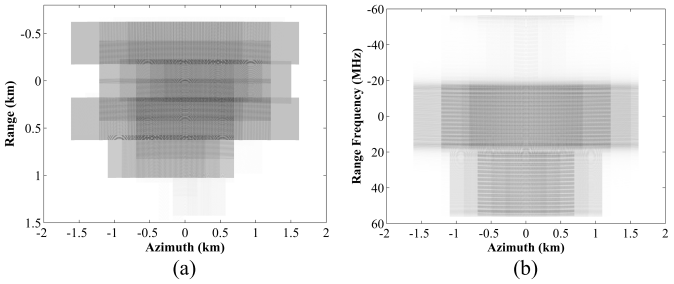

Fig. 10. Received echoes in (a) range-azimuth time domain and (b) range frequency-azimuth time domain.

rationally designed NSPCs. Eight stationary point targets and one moving target are individually arranged in each area, and the positions of the targets set in the three regions are given in Fig. 9(a)–(c), where each region differs by a maximum unambiguous range. The whole processing of echo subpulses separation and range ambiguity suppression will be described in detail.

Fig. 10 shows the received echoes in the range-azimuth time domain and the range frequency-azimuth time domain, respectively. Fig. 10(a) illustrates clearly that the echoes from three subpulses are aliased in the time domain, and Fig. 10(b) depicts the subpulses occupying three different frequency bands which can evidently be distinguished. Fig. 11(a) gives the range compressed results of these range ambiguous echoes, and Fig. 11(b) is the azimuth slice of Fig. 11(a). In addition, the imaging results for all the ambiguous echoes are shown in Fig. 11(c). It is apparent that the targets in the first and second regions are seriously aliased, and the second region targets are only −2 dB lower than those in the first. For the NSPC-SAR system, overlapping echoes will first be differentiated in the frequency domain. It is noted that the sidelobes of other subpulses in the frequency domain also overlap and cause mutual interference. Then, the first subpulse parameters are taken for down-conversion and bandpass filtering, as shown in Fig. 12. Fig. 12(a) reveals that the echoes are distinguished in the time domain without aliasing, and Fig. 12(b) is the extraction result for the first subpulse with bandpass filtering, which completes the separation of echoes in the frequency domain.

Fig. 13 explicates the range-compressed outcome of the first subpulse, where Fig. 13(a) is in the range-azimuth time domain, and Fig. 13(b) is the azimuth slice of Fig. 13(a). It is obvious that the residual range-ambiguous energies from side-lobe still exist, which are about −20 dB. Therefore, we will continue to suppress the residual range ambiguities using DBF technology and the data reconstruction method, which means better to eliminate the remaining interference with the information of the second and third subpulses. Fig. 14 presents the results of residual range ambiguity suppression for the first subpulse. By comparing Figs. 13(a) and 14(a), and 13(b) and 14(b), only the targets in the first region exist, indicating that the residual range ambiguity has been thoroughly

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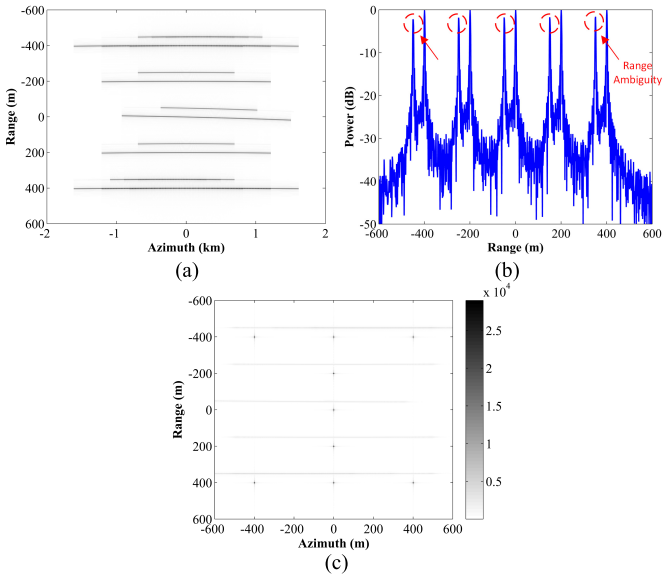

Fig. 11. Range compressed results before range ambiguity suppression. (a) Range compressed result. (b) Azimuth slice of (a). (c) Imaging result.

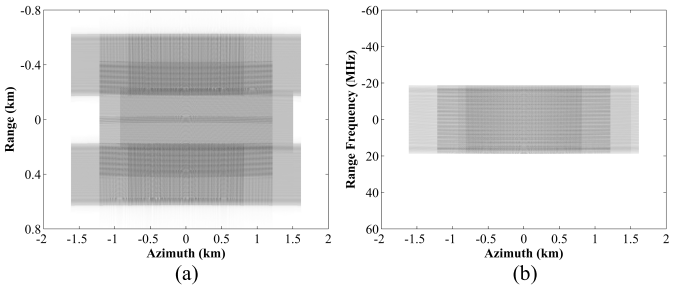

Fig. 12. Distinguished first subpulse echo in (a) range-azimuth time domain and (b) range frequency-azimuth time domain.

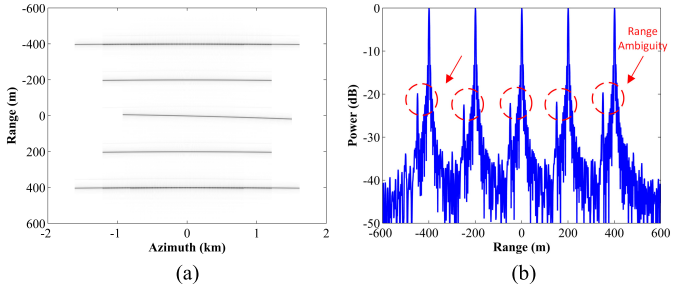

Fig. 13. Range compressed results of the first subpulse. (a) In the range-azimuth time domain. (b) Azimuth slice of (a).

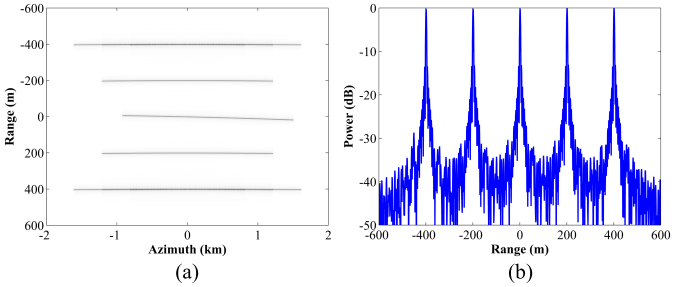

Fig. 14. Range compressed results of the first subpulse after residual range ambiguity suppression. (a) In the range-azimuth time domain. (b) Azimuth slice of (a).

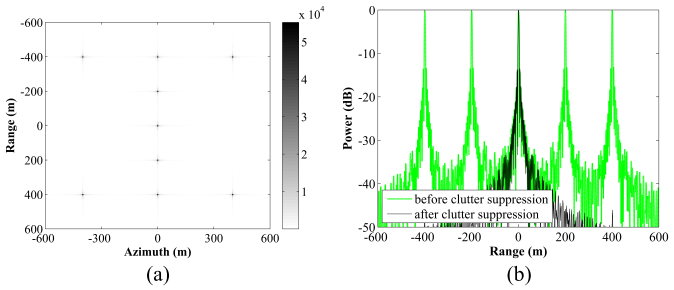

Fig. 15. Processing results of the first region. (a) Imaging result. (b) Clutter suppression result.

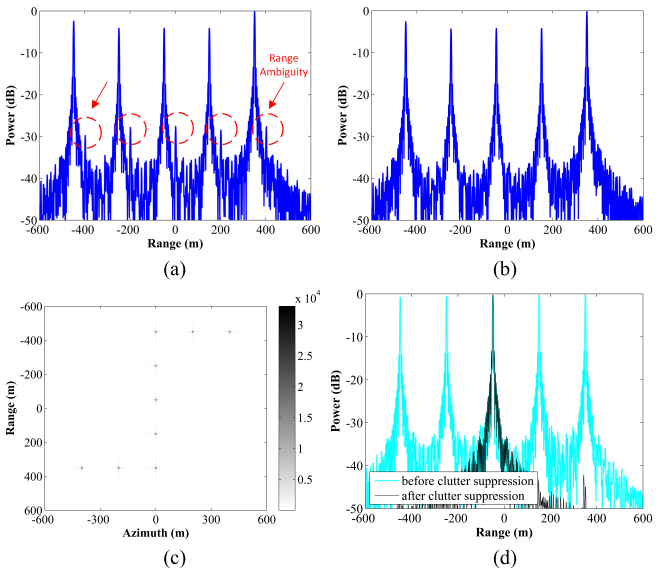

Fig. 16. Processing results for the second region. (a) Azimuth slice after first range ambiguity suppression. (b) Azimuth slice after second range ambiguity suppression. (c) Imaging result. (d) Clutter suppression result.

suppressed and the interferences from other subpulses have been removed. That is to say, the NSPC scheme can effectively suppress the range ambiguity through the frequency-domain separation and DBF technique, which is conducive to wide-swath imaging and successive moving target detection.

After suppressing the ambiguous energies from the main-lobe and sidelobe, the imaging result of the first region without ambiguities is shown in Fig. 15(a). Then, Fig. 15(b) gives the clutter suppression results in the first region. It can be seen that only the moving target exists, while the targets in other regions and the stationary targets from the first region are well suppressed, and the clutter energy is suppressed to below −45 dB. Similarly, Fig. 16 demonstrates the processing flow of the second region. Fig. 16(a) is the slice of the range compression result for the second subpulse, and the remaining range ambiguity can also be observed. In contrast, Fig. 16(b) shows the slice result after the residual range ambiguity suppression, and there is no interference of the range ambiguity at this time. Fig. 16(c) is the imaging of

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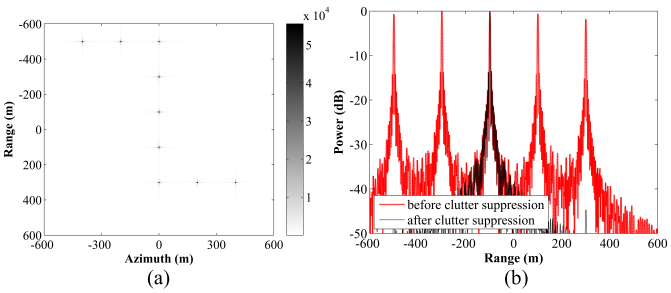

Fig. 17. Processing results for the third region. (a) Imaging result. (b) Clutter suppression result.

TABLE IV
PSLR AND ISLR OF THREE REGIONS

Parameters	Region 1		Region 2		Region 3	
	PSLR	ISLR	PSLR	ISLR	PSLR	ISLR
***Azimuth*(dB)**	-13.26	-10.62	-13.25	-10.60	-13.22	-10.54
***Range*(dB)**	-12.99	-10.35	-13.20	-10.43	-13.10	-9.47

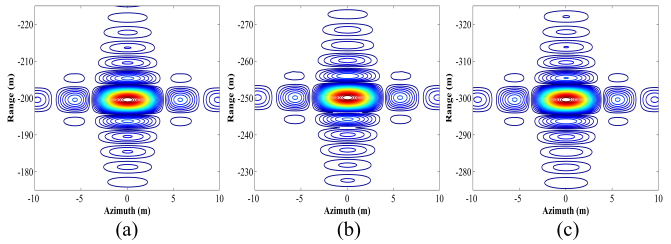

Fig. 18. 2-D sectional views of the stationary target imaging result from three regions. (a) First region. (b) Second region. (c) Third region.

the targets in the second region, and Fig. 16(d) is the result of clutter suppression, which can detect the targets without ambiguity. All the above signify that the targets in its own region are well focused and there is no interference of the range ambiguity from other regions. In the same way, the range ambiguity suppression results of the third region can be obtained. Fig. 17 depicts the processing in the third region.

In order to give the focusing performance of the NSPC method more intuitively, we take the peak to sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) of the point target to evaluate it. Fig. 18(a)–(c) shows the 2-D sectional views after interpolation of the stationary target imaging result selected from three regions, respectively. The PSLR and ISLR in the range and azimuth dimensions of each region are listed in Table IV, which reveals that the point targets have achieved well focusing. In this article, we quantify the range ambiguity suppression capability by the metric range ambiguity-to-signal ratio (RASR) [56]. The RASR means the ratio of the sum of the range ambiguity power to the desired signal power. In Fig. 19(a)–(c), they manifest the RASR variation curves along the slant-range corresponding to the three imaging regions individually. Among them, Fig. 19(a) gives a comparison between the traditional imaging method before range ambiguity suppression and the NSPC method

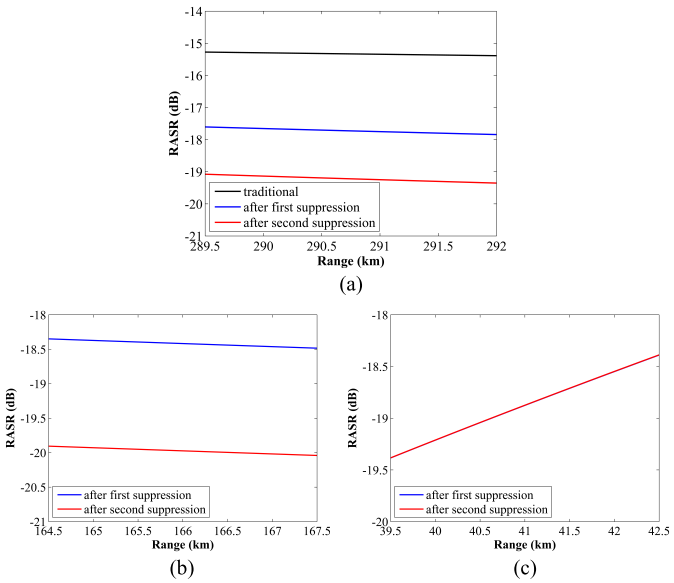

Fig. 19. RASR after the first and the second suppression of three regions. (a) Comparison between the traditional method and NSPC method of the first region. (b) Second region. (c) Third region.

in this article. It can be observed that the RASR of the traditional method is about −15.3 dB. Meanwhile, the RASR of the NSPC method is about −17.6 dB after the first range ambiguity suppression, and it is all lower than −19 dB after the second suppression. The performance advantage of the proposed method for range ambiguity suppression is demonstrated with evidence. In addition, the power of the range ambiguity mainly depends on the corresponding pattern gain. The sidelobe pattern corresponding to the ambiguity is in a downward trend from the near-end to the far-end, which is consistent with the curve. Likewise, Fig. 19(b) illustrates the RASR of the second region after two suppressions. As for the third region, since the first and second regions are far apart from the third, the ambiguities of the two entering from the sidelobes are very small. Therefore, the curves after the first and second suppressions almost coincide, as shown in Fig. 19(c).

For the point targets, the simulation shows in detail that the NSPC method can guide the beam by designing parameters to the desired direction, and can effectively exploit the frequency band resources to realize the separation of multiple subpulses. Also, it can achieve the auto-scanning of the beams and increase the imaging areas in elevation as required. In addition, the information of the relevant subpulses causing the range ambiguity is fully utilized to suppress the interference corresponding to the subpulse that should be extracted. Consequently, the increase of the false alarm probability caused by the moving targets of range ambiguity from other regions can be further eliminated.

B. Simulation of Distributed Targets for NSPC-SAR

In this section, we will demonstrate the ability of the NSPC method to suppress the range ambiguity of the distributed targets. Three subpulses are set reasonably to point

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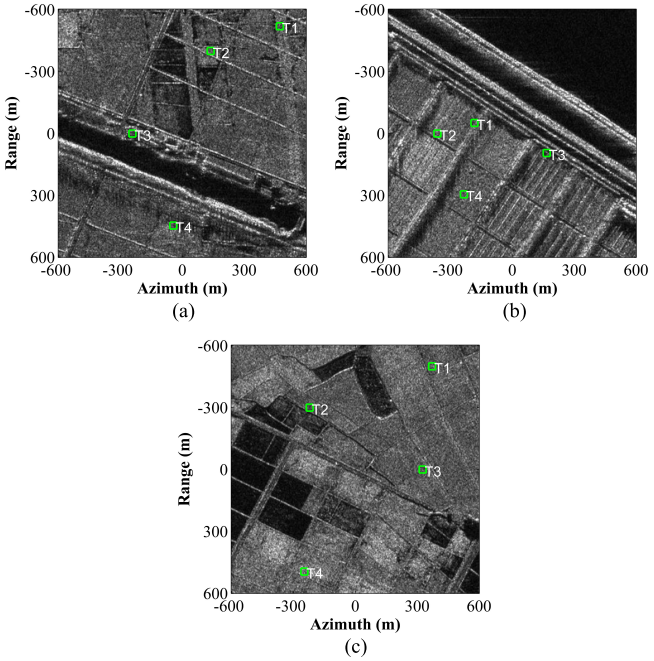

Fig. 20. Original scene images of three regions. (a) First region. (b) Second region. (c) Third region.

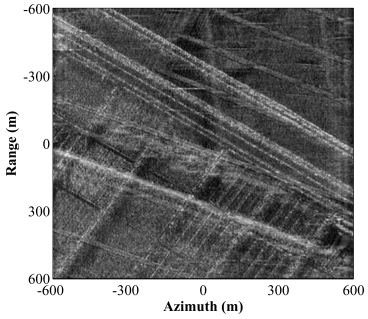

Fig. 21. Imaging result of ambiguous echoes.

to the three range ambiguous regions, respectively, where there are no echoes being blocked. In this simulation, the parameter settings of the three regions are the same as in Table III.

First, the original scene images of the three regions are presented, as shown in Fig. 20(a)–(c), in which the moving targets' positions are marked, respectively. With range ambiguities, the focusing result for the second range region is shown in Fig. 21. It is obvious that the first region is aliased with the imaging scene, and at this time the ambiguous interferences from stationary clutter will be unfavorable for moving target detection.

Under the strategy of this article, the entire processing flow of distributed targets is the same as that of the point targets. First, the subpulses are separated in the frequency domain, and then the secondary range ambiguity suppression is performed. Finally, the imaging results of three regions without ambiguity can be obtained, as shown in Fig. 22(a)–(c). Next, moving target detection is conducted on the imaging results, respectively, as shown in Fig. 22(d)–(f). The GoDec

TABLE V
TARGET VELOCITIES ESTIMATION RESULTS

Region 1	T1	T2	T3	T4
Measured Value (m/s)	0.96	0.43	-0.66	0.57
True Value (m/s)	1.00	0.40	-0.70	0.60
Region 2	**T1**	**T2**	**T3**	**T4**
Measured Value (m/s)	0.73	-1.45	0.62	-2.56
True Value (m/s)	0.70	-1.50	0.60	-2.60
Region 3	**T1**	**T2**	**T3**	**T4**
Measured Value (m/s)	2.98	-4.35	5.54	-3.76
True Value (m/s)	3.00	-4.40	5.60	-3.80

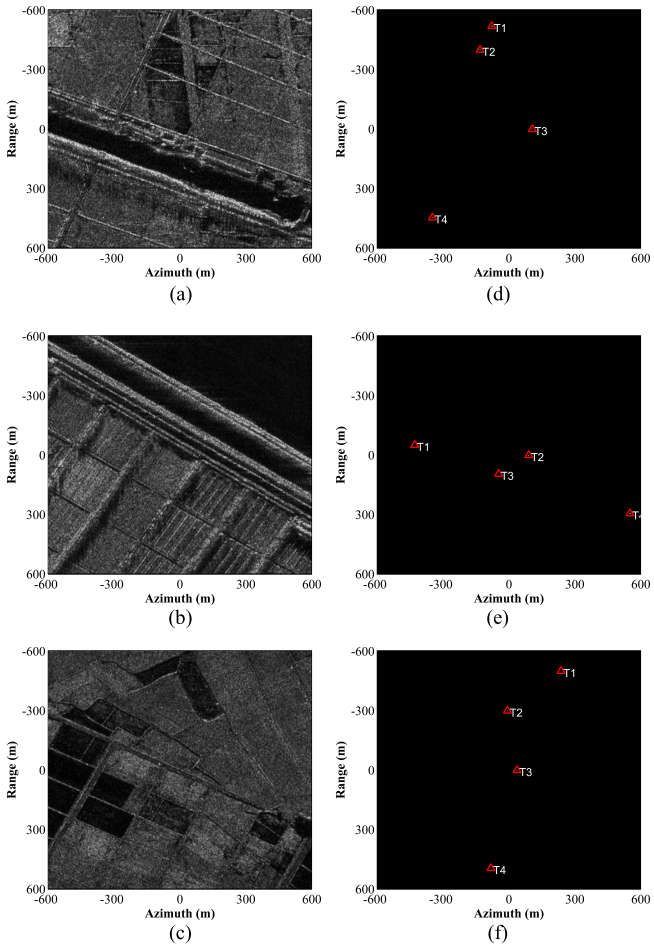

Fig. 22. Imaging results for (a) first region, (b) second region, and (c) third region. Moving targets detection by GoDec algorithm for (d) first region, (e) second region, and (f) third region.

algorithm in Table I is used to separate the moving targets and clutter, and the detected moving targets are marked with red triangles in Fig. 22(d)–(f). The velocities of the targets lead to deviation of the positions during imaging. Fig. 23(a)–(c) shows the detected positions and relocations of the moving targets. In Fig. 23(a)–(c), the red triangles are the detected positions of the moving targets, and the green squares are the positions of the relocated targets according to the measured

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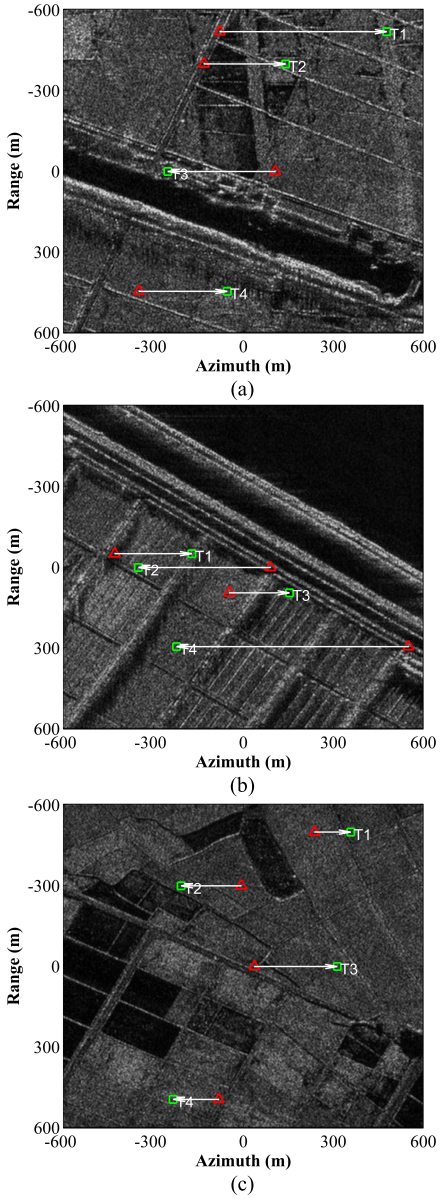

Fig. 23. Detected moving targets' position and relocation results for (a) first region, (b) second region, and (c) third region.

velocities. Besides, the true and the measured velocities of the moving targets in the three regions are shown in Table V. The display of the result implies the accuracy of velocity measurement without ambiguity is relatively high. Distributed targets simulation illustrates that the proposed method in this article can specify the orientation of the scanning beams in elevation, further improve the imaging width and detect moving targets without ambiguity. Also, the method of removing range ambiguity is uncomplicated and practically feasible.

VI. Conclusion

In this article, a novel NSPC-SAR system is established to solve range ambiguity. The separation of echoes and the suppression of range ambiguity are realized by using the frequency band resources of the transmitter and the loaded subpulses coding. Through the targeted NSPCs, the specific directions of the subpulses can be set while the elevation beams are automatically scanned. It can make the beams skip the invalid blind areas where the echoes are blocked, which can save time and improve the utilization of energy resources. In addition, by designing subpulses occupying different frequency bands, the frequency domain bandpass filter, and DBF technology are used to separate the subpulses and eliminate the interference between the mutual subpulses, which effectively realizes the range ambiguity suppression. Furthermore, simulations of point targets and distributed targets verify the effectiveness of the proposed method, which can greatly improve the imaging width and the accuracy of subsequent moving target detection.

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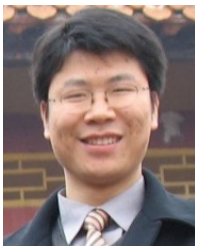

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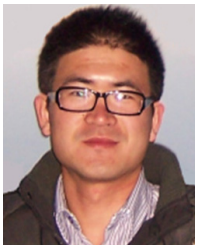

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